# flucut

INFLUENZA A AMPLICON QC, BEFORE IRMA

# Primers off, the internal-priming artefact caught.

flucut removes the universal RT-PCR primers (Uni12/Uni13) from influenza A reads and, with references, finds and measures **internal priming** — the hidden artefact that writes wrong bases into the consensus (the HA I348K lysine). One command before IRMA, no configuration, on Nanopore and Illumina.

**1.6 s**
for 32,000 reads
4 threads, libdeflate

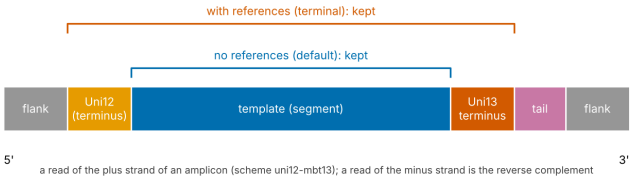
**60 MB**
of memory
runs on any laptop

**99.6%**
internally primed reads tolerated, still no N in the consensus
~91% for the IRMA-core trimmer

**8 / 8**
segments classified, terminal vs internal, by strand
reported, not trimmed silently

HOW IT WORKS

## One read, one amplicon



a read of the plus strand of an amplicon (scheme uni12-mbt13); a read of the minus strand is the reverse complement

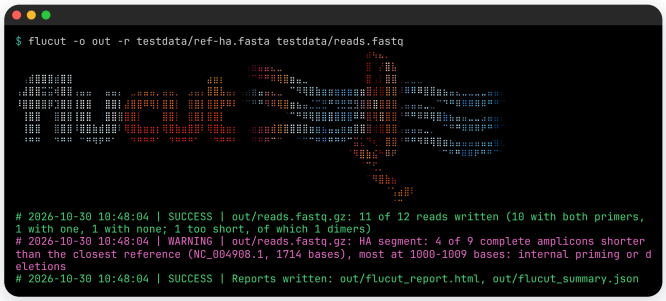
- the primer is searched only near the read end, aligned with errors, on both strands, degenerate bases and all
- the sequence after the primer (a probe) is located in the segment: at the terminus it is a real amplicon, deeper inside it is internally primed
- an internal or unsure end loses its **whole** primer (cut when unsure), so the artefact never reaches the consensus
- Nanopore adapters (native, rapid, ligation) removed at broken ends; one IRMA-ready FASTQ per sample

BUILT FOR THE PROTOCOL

## No primer files, no tuning

**Uni12 / Uni13** | mbt tailed | uni | Nanopore | Illumina | FASTQ / BAM | MinKNOW dirs

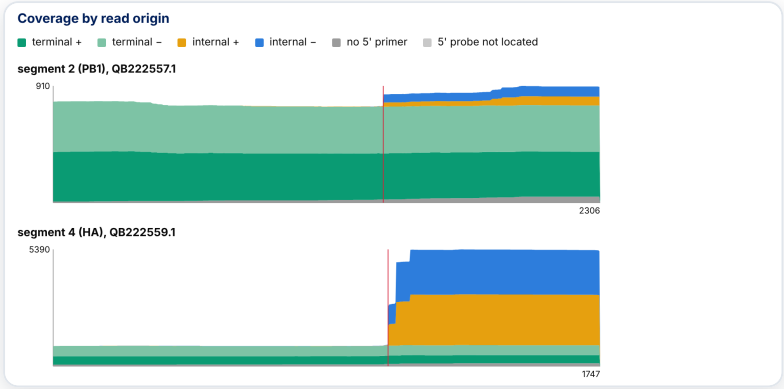
- the schemes and primers are built in — no `-g/-a/-e` to guess
- the error rate is measured on the first reads ( `-e auto` ); a Nanopore and an Illumina profile ( `-p` )
- with `-r` , the conserved 12/13-base termini are kept at the ends confidently terminal — complete segment ends in one pass



WHAT THE OTHERS MISS

# See the internal priming — don't just trim it

Every primer-aware trimmer removes the Uni12 footprint. Only flucut tells you it was there: it reconstructs the coverage of each segment from its own read classification, so internal priming shows as a step up above the full-length baseline — here the HA I348K site, in a real H5N1 Nanopore run.




**Coverage by read origin** (SRA SRR36760543, complete H5N1 segment references). Green: full-length amplicons, a flat baseline. Amber + blue: internally primed reads, by strand, beginning at the red site line. HA steps up sharply (the I348K artefact); PB1 shows a smaller internal site.

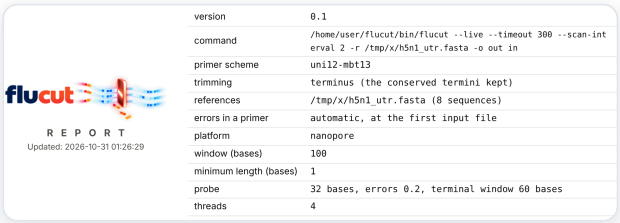
| | flucut | cutadapt | BBDuk | IRMA-core | dorado | Barbell |
|---|---|---|---|---|---|---|
| Nanopore | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Illumina | ✓ | ✓ | ✓ | ✓ | – | – |
| Max internal priming, no N in consensus† | 99.6% | 99.6%‡ | 93% | 91% | 99.7% | – |
| Keeps the correct segment ends in the consensus | ✓ | – | – | – | – | – |
| Internal-priming analysis + coverage plot | ✓ | – | – | – | – | – |
| Built-in flu protocol, no primer file | ✓ | – | – | ✓ | – | – |
| Speed / memory, 32k reads | 1.6 s / 60 MB | 3.7 s / 41 MB | 4.0 s / 950 MB | 7.5 s / 11 MB | 8.1 s / 480 MB | 2.4 s / 44 MB |

† the highest share of internally primed reads at the HA site at which IRMA still calls the correct base; derived from the measured read retention ( `benchmark/COMPARISON.md` ). ‡ cutadapt reaches it only by also discarding ~27% of the genuine full-length reads. Barbell removes barcodes, not the internal Uni primer, so it cannot correct the artefact.

REAL TIME

## Live, while you sequence

With `--live`, flucut follows a MinKNOW run: each new file is trimmed as it is written, and the report and JSON summary are updated and reloaded in the browser — the internal priming of the run building up live, on the sequencing laptop. Resume, interruption and end-of-run are handled, and the output matches a normal run on the final files.




GET FLUCUT

## Free and open source

**github.com/baia-ipc/flucut**

C, ISC license, Linux and macOS, no dependencies.

```
$ make
$ flucut -r refs.fasta -o out/ \
   run/fastq_pass/
$ IRMA FLU-minion out/*.fastq.gz sample
```